physics and special divine action. They have ignored the fact that Newtonian laws are stated for *causally closed* systems, but obviously no system in which God acts specially would be a closed system; hence God's special action would not go contrary to the laws of Newtonian physics. And with the advent of quantum mechanics it has become harder yet to find conflict between special divine action and current physics.

There are other areas of science, however, where the appearance of conflict is matched by reality. I'm thinking in particular of *evolutionary psychology* and *scientific scripture scholarship*—historical Biblical criticism, as it is sometimes called. You may think that historical Biblical criticism and evolutionary psychology have little in common. Perhaps so; still, they are alike in a couple of ways crucial to our inquiry. First, in each of these areas we find claims and assertions incompatible with theistic or Christian belief; here there really is conflict. But second, I call this conflict "superficial," and in chapter 6 I will explains why. Very briefly, however: even though Christians are committed to a high view of science, and even if these disciplines do constitute science or good science (a state of affairs that is by no means self-evident), these developments in evolutionary psychology and historical Biblical criticism don't offer, or even threaten to offer, *defeaters* for Christian or theistic belief. Hence there is conflict, but it is merely superficial.

I EVOLUTIONARY PSYCHOLOGY

Those who accept Darwinism, as one might expect, often try to explain various features of contemporary organisms in Darwinian terms. One way to do this is to show how the trait in question does or did contribute to the reproductive fitness of the organisms in question. Why do tigers have stripes? Because they provide camou-

flage, which, given the tiger's lifestyle, is fitness-enhancing. Or perhaps the trait in question is not itself adaptive, doesn't itself contribute to fitness, but is genetically associated with traits that do. As a special case, perhaps the trait in question is a spandrel, a trait that isn't itself fitness-enhancing, but is a consequence of traits that are.[1] Such explanations sometimes seem plausible, at least within limits: if you think the organisms in question arose by virtue of natural selection, then this seems a fairly sensible way in which to try to explain at least some of their traits.

As one might expect, this sort of explanation has been extended to human beings as well; one enterprise of giving such explanations is sociobiology, or, as it is now called, evolutionary psychology. Sociobiology burst upon the scene in 1975 with the publication of E. O. Wilson's *Sociobiology: The New Synthesis*. Wilson's work provoked pointed criticism. Some claimed it tended to dismiss rape and other violence as merely a part of our evolutionary heritage; others claimed it was deeply sexist; still others declared it bad science. Work along these lines is now called "evolutionary psychology"; we need not enter the controversy as to whether evolutionary psychology is a successor discipline to sociobiology, or just the same thing renamed to deflect criticism. Either way, evolutionary psychology has been growing, and is rapidly becoming an established part of academic psychology.

In essence, evolutionary psychology is an attempt to explain important human traits and behaviors in terms of the evolutionary origin of the human species. The heart and soul of this project is the effort to explain distinctive human traits—our art, humor, play, love, sexual behavior,

1. Stephen Jay Gould and Richard Lewontin: "Spandrels—the tapering triangular spaces formed by the intersection of two rounded arches at right angles—are necessary architectural byproducts of mounting a dome on rounded arches." "The Spandrels of San Marco and the Panglossian Paradigm: a Critique of the Adaptionist Program" (London: *Proceedings of the Royal Society*, B 205, 1979), p. 581.

poetry, sense of adventure, love of stories, our music, our morality, and our religion itself—in terms of adaptive advantages accruing to our hunter-gatherer ancestors back there on the plains of Serengeti.

Some of this can be a pretty tall order. Take our love of beauty, for example: here it isn't easy to see what an evolutionary explanation would look like. (Of course it is always possible to declare a trait a spandrel.) There is the glorious beauty and grandeur of mountains—Mt. Baker, for example, or Mt. Shuksan, or the Grand Teton, or any of a hundred more. There is the splendor of a craggy ocean shore, but also of a tiny highly articulate flower. Alan Shepard, the first American in space, gasped at the sheer beauty of the earth as seen from space. It is hard to see how a capacity to find marvelous beauty in such things would be of adaptive use to our hunter-gatherer ancestors. And of course there is Mozart, and Bach. Perhaps we can see how love of something like heavy metal rock could be adaptively useful, possibly like the martial airs that encourage troops going into battle. But Mozart's Ave Verum Corpus? Bach's B-minor Mass?

Harvard evolutionary psychologist Steven Pinker once told a gathering of musicologists why music had rated only eleven pages of his 660-page *How the Mind Works*: "He told the musicologists why the topic did not merit more attention: music was 'useless' in terms of human evolution and development. He dismissed it as 'auditory cheesecake,' a trivial amusement that 'just happens to tickle several important parts of the brain in a highly pleasurable way, as cheesecake tickles the palate.'"[2] Anthropologist Steven Mither defends music against this charge of being useless: music does too have evolutionary significance, he said, because it is connected with walking and marching and other rhythmical activities.[3] It's a little startling to

2. Pinker, *How the Mind Works* (New York: Penguin, 1997). Pinker's statement as reported by Rodney Clapp in *The Christian Century*, February 10, 2009, p. 45.
3. Mither, *The Singing Neanderthals* (London: Weidenfeld and Nicolson, 2005).

see something as deep, powerful, and significant as music denigrated or defended in those terms. Is an activity important only if it has played a prominent role in our evolution, enabling our ancestors to survive and reproduce? What about physics, mathematics, philosophy, and evolutionary biology itself: do (did) they have evolutionary significance? After all, it is only the occasional assistant professor of mathematics or logic that needs to be able to prove Gödel's theorem in order to survive and reproduce. Indeed, given the nerdness factor, undue interest in such things would have been counterproductive in the Pleistocene. What prehistoric woman would be interested in some guy who prefers thinking about set theory to hunting?

Evolutionary psychology is extremely popular, at present, and growing in popularity. As I said, it is becoming an established part of academic psychology; it is also gaining purchase in more popular culture. Just about every other issue of *The New York Review of Books* carries a review of still another book intended to interpret ourselves to ourselves along these lines. A recent high (or maybe low) point is a book in which a new understanding of *religion* is proposed. At a certain stage in our evolutionary history, so the claim goes, we human beings made the transition from being prey to being predators. Naturally that occasioned great joy, and religion arose as a celebration of that happy moment! Granted, that sounds a little far-fetched: wouldn't we have needed the consolations of religion even more when we were still prey? Still, that was the claim. Some of these projects, therefore, are a little hard to take seriously, but others are both intellectually challenging and adorned with all the trappings of serious science, complete with mathematics, models, the fitting of models to data, and the sort of stiff, impersonal literary style in which science is properly written.

Even a cursory glance at the literature shows that many theories from this area of science seem, at least on first inspection, to be deeply problematic from a Christian perspective. According to Michael Ruse

and E. O. Wilson, "ethics is an illusion fobbed off on us by our own genes to get us to co-operate; thus morality ultimately seems to be about self-interest."[4] They also claim that "humans function better if they are deceived by their genes into thinking there is a disinterested objective morality binding upon them, which we should obey."[5] Why so? Individuals with our moral intuitions will be likely to cooperate with each other; groups with our moral intuitions will therefore do better, from the point of view of survival and reproduction, than groups that lack those intuitions.[6] What has been selected for, then, are people with a penchant for forming a twofold belief on this head. First, they have the sense that there really is such a thing as objective obligation, a categorical *ought* that holds, whatever your goals or aims, whatever you or anyone thinks or desires. And second, they think that among these obligations is something like the golden rule: treat others (or at least others in your group) as you yourself would like to be treated. We are thus inclined to think that this injunction is an objective and categorical requirement of morality. According to Ruse and Wilson, however, our thinking this is no more than a trick played on us by our genes to get us to cooperate: in fact there aren't any such objective moral obligations or requirements.

Consider an example. Herbert Simon's article "A Mechanism for Social Selection and Successful Altruism" is concerned with the problem of altruistic behavior—that is, behavior that promotes the fitness of someone else at the expense of the altruist's own fitness.[7]

4. Ruse and Wilson, "The Evolution of Ethics," in *Religion and the Natural Sciences: The Range of Engagement*, ed. James Huchingson (San Diego: Harcourt Brace, 1993), p. 310.
5. Ruse and Wilson, "Moral Philosophy as Applied Science," *Philosophy* 61, 1986, p. 179.
6. There has been controversy about whether the notion of group selection is viable; for a spirited and convincing argument that it is, see Elliott Sober and David Sloan Wilson, *Unto Others: The Evolution and Psychology of Unselfish Behavior* (Cambridge: Harvard University Press, 1998).
7. Simon, *Science* vol. 250 (December, 1990) pp. 1665ff. Simon won a Nobel Prize in economics, but later became professor of computer studies and psychology at Carnegie-Mellon University in Pittsburgh.

Why, asks Simon, do people like Mother Teresa, or the Scottish missionary Eric Liddel, or the Little Sisters of the Poor, or the Jesuit missionaries of the seventeenth century—why do these people do the things that they do? Why do they devote their time and energy and indeed their entire lives to the welfare of other people? Of course it isn't only the great saints of the world that display this impulse; most of us do so to one degree or another. Most of us give money to help feed and clothe people we have never met; we may support missionaries in foreign countries; we try, perhaps in feckless and fumbling ways, to do what we can to help the widow and orphan. Now how, asks Simon, can we account for this kind of behavior? Given our evolutionary origin in natural selection, a rational, properly functioning human being could be expected to act or try to act in such a way as to increase her personal fitness—that is, to act so as to increase the probability that her genes will be widely disseminated in the next and subsequent generation, thus doing well in the evolutionary derby.[8] A paradigm of rational behavior, so conceived, was reported in 1991: "Cecil B. Jacobson, an infertility specialist [in Alexandria, Virginia], was accused of using his own sperm to impregnate his patients; he may have fathered as many as 75 children, a prosecutor said Friday."[9] Unlike Jacobson, however, such people as Mother Teresa and Thomas Aquinas cheerfully ignore the short or long term fate of their genes. What is the explanation of this behavior?

This behavior, says Simon, is to be explained at the individual level in terms of two mechanisms: "docility" and "bounded rationality":

> Docile persons tend to learn and believe what they perceive others in the society want them to learn and believe. Thus the

8. More simply, says Simon, "Fitness simply means expected number of progeny" (p. 1665).
9. *The South Bend Tribune*, December 21, 1991.

content of what is learned will not be fully screened for its contribution to personal fitness.

> Because of bounded rationality, the docile individual will often be unable to distinguish socially prescribed behavior that contributes to fitness from altruistic behavior. In fact, docility will reduce the inclination to evaluate independently the contributions of behavior to fitness.....By virtue of bounded rationality, the docile person cannot acquire the personally advantageous learning that provides the increment, *d*, of fitness without acquiring also the altruistic behaviors that cost the decrement, *c*.[10]

The idea is that a Mother Teresa or a Thomas Aquinas display "bounded rationality"; they are unable to distinguish socially prescribed behavior that contributes to fitness from altruistic behavior (socially prescribed behavior which does not). As a result they fail to acquire the personally advantageous learning that provides that increment *d* of fitness without, sadly enough, suffering that decrement *c* exacted by altruistic behavior. They acquiesce unthinkingly in what society tells them is the right way to behave; and they aren't quite up to making their own independent evaluation of the likely bearing of such behavior on the fate of their genes. If they *did* make such an independent evaluation (and were rational enough to avoid silly mistakes), they would presumably see that this sort of behavior does not contribute to personal fitness, drop it like a hot potato, and get right to work on their expected number of progeny. From a Christian perspective, obviously, this explanation of the behavior of Mother Teresa and other altruists is wildly off the mark—not even close enough to be a miss.

10. That is (in Simon's sense), behavior that exacts a cost in terms of the agent's fitness. "A Mechanism for Social Selection and Successful Altruism," pp. 1666–67.

II EVOLUTIONARY PSYCHOLOGY AND RELIGION

I suppose the place to look for the most overt conflict would be the accounts of religion itself that are proffered by the scientific study of religion, including evolutionary psychology and cognitive science. Of course many of these theories of religion are incompatible with each other: according to some, religion is adaptive, but according to others it is nonadaptive or even maladaptive (a malignant virus, as Richard Dawkins seems to think); according to some, religion arises by way of group selection, but according to others group selection is impossible. Still, there is no dearth of examples of proclamations that conflict with religious belief (not to mention with each other). For example, Steven Pinker asks: "How does religion fit into a mind that one might have thought was designed to reject the palpably not true?" and states that "religion is a desperate measure that people resort to when the stakes are high and they have exhausted the usual techniques for the causation of success—medicines, strategies, courtship, and, in the case of the weather, nothing." He goes on, resurrecting the old canard about how religion is the result of crafty priests and credulous parishioners: "I have alluded to one possibility: the demand for miracles creates a market that would-be priests compete in, and they can succeed by exploiting people's dependence on experts. I trust such experts as dentists and doctors; that same trust would have made me submit to medical quackery a century ago and to a witch doctor's charms millennia ago."[11]

This is perhaps more a declaration of personal dislike for religion than a scientific or semi-scientific pronouncement, but there are many others. Rodney Stark proposed a theory according to which religion is a kind of spandrel of rational thought, an attempt to acquire nonexistent goods—eternal life, a right relationship with

11. Pinker, *How the Mind Works*, pp. 554, 556, 557.

God, salvation, remission of sins—by negotiating with nonexistent supernatural beings.[12] The idea is that rational thought, that is, means/ends or cost/benefit thinking, comes to be in the usual evolutionary way. But having the capacity for such thought inevitably carries with it the capacity to pursue nonexistent goals, like the pot of gold at the end of the rainbow, or the ones connected with religion. Taken neat, this theory is clearly incompatible with Christian belief, according to which at least some of the supernatural beings and some of the goods mentioned do indeed exist. Another example: David Sloan Wilson (not to be confused with E. O. Wilson) suggests that religion is essentially a means of social control employing or involving fictitious belief.[13] Again, taken neat, this is incompatible with Christian belief.

Among the most important writers on the scientific study of religion would be Pascal Boyer, Scott Atran, and Justin Barrett ; there is also a popular presentation of some of this material in Daniel Dennett's *Breaking the Spell*.[14] Like the early Stark, Boyer takes religion to be a kind of spandrel: it is a consequence or side effect of having the sort of brain we actually have. That we have that sort of brain is to be explained in the usual way, in terms of its contribution

12. The views of the "old" Rodney Stark must be carefully distinguished from those of the "new," which are vastly more friendly to religion: see, e.g., (with Roger Finke) *The Churching of America* (New Brunswick: Rutgers University Press, 1992), *The Rise Of Christianity* (Princeton: Princeton University Press, 1996), and *Discovering God* (New York: Harper Collins, 2007). As David Sloan Wilson puts it, "[For Stark] Religion is envisioned as an economic exchange between people and imagined supernatural agents for goods that are scarce (e.g., rain during a drought) or impossible (e.g. immortal life) to obtain in the real world" (*Darwin's Cathedral: Evolution, Religion, and the Nature of Society*, Chicago. University of Chicago Press, 2002, p. 52). Stark has since rejected this theory.
13. *Darwin's Cathedral: Evolution, Religion, and the Nature of Society*. I'll say more about Wilson's theory later.
14. Boyer, *Religion Explained* (New York: Basic, 2001); Atran, *In Gods We Trust* (Oxford: Oxford University press, 2002); see also Todd Tremlin, *Minds and Gods* (Oxford: Oxford University Press, 2006), and D. Jason Slone, *Theological Incorrectness: Why Religious People Believe What They Shouldn't* (New York: Oxford University Press, 2004); Barrett, "Exploring the Natural Foundations of Religion," in *Trends in Cognitive Sciences*, vol. 4, 2000, pp. 29–34; and Dennett, *Breaking the Spell* (New York: Penguin Books, 2006).

to fitness; but given such a brain, according to Boyer, the development of religion naturally occurs. In essence, he thinks, religion is a whole family of cognitive phenomena involving "counterintuitive" beings (beings who act in ways counter to our ordinary categories): for example, religion often involves beings who can act in the world without being visible. This out-of-the-ordinary counterintuitive character of the beings in question attracts attention and makes them memorable. However they must be *minimally* counterintuitive; too much departure from the ordinary makes them both hard to take seriously and hard to remember. A box of Kleenex that is secretly the ruler of the world, for example, is *too* counterintuitive; a religion of which this was the central feature probably wouldn't thrive.

Atran endorses several features of Boyer's theory and adds more: according to him "religion is (1) a community's costly and hard-to-fake commitment (2) to a counterfactual and counterintuitive world of supernatural agents (3) who master people's existential anxieties such as death and deception."[15] According to Atran, therefore, not just any old counterfactual and counterintuitive idea will make a religion (a whistling teakettle orbiting the sun, for example, won't fill the bill); the ideas have to involve supernatural *agents*. And not just any old remarkable agent will fill the bill either (Mickey Mouse, he says, will not); the agents have to address and assuage people's existential anxieties. So why is it that human beings are religious? Like Boyer, Atran believes that while our minds have come to be and have been developed by natural selection, religion itself isn't adaptive, but is a byproduct of our cognitive architecture:

> Religion is materially expensive and unrelentingly counterfactual and even counterintuitive. Religious practice is costly in terms of material sacrifice (at least one's prayer time), emotional

15. Atran, *In Gods We Trust*, p. 4.

expenditure (inciting fears and hopes) and cognitive effort (maintaining both factual and counterintuitive networks of belief).[16]

Now some writers seem to think that in coming up with a suggestion as to the evolutionary origin of religion, they are in some way discrediting it.[17] Apart from that gratuitous "counterfactual," however, there is nothing in Boyer or Atran that is inconsistent with theistic or Christian belief (although both seem at best extremely skeptical of such belief). Describing the origin of religious belief and the cognitive mechanisms involved does nothing, so far, to impugn its truth.[18] No one thinks describing the mechanisms involved in perception impugns the truth of perceptual beliefs; why should one think things are different with respect to religion? According to Christian belief, God has created us in such a way that we can know and be in fellowship with him. He could have done this in many ways; for example, he could have brought it about that our cognitive faculties evolve by natural selection, and evolve in such a way that it is natural for us to form beliefs about the supernatural in general and God himself in particular. Finding a "natural" origin for religion in no way discredits it.[19]

16. Atran, *In Gods We Trust*, p. 4.
17. For example, according to Jesse Bering, director of the Institute of Cognition and Culture, Queens University, Belfast, who apparently believes that with his research, "we've got God by the throat and I'm not going to stop until one of us is dead" ("The God Fossil," *Broward Palm Beach New Times*, March 9, 2006). (Do you suppose God is trembling in the face of this threat?)
18. Many writers seem not to grasp this obvious fact. The same goes for suggestions as to what components and dynamics of the brain underlie religious experience, as in Michael Persinger, "The sensed presence within experimental settings: Implications for the male and female concept of self," *Journal of Psychology: Interdisciplinary and Applied* 137:1 (2003). Suppose centers for perception were identified in the brain, and suppose artificial stimulation of these centers could cause experiences as of seeing a tree; would this cast doubt upon our perception of trees?
19. See Kelly Clark and Justin Barrett, "Reformed Epistemology and the Cognitive Science of Religion." *Faith and Philosophy* 28 (2010), pp. 174–89.

A more promising line of criticism: it might be suggested that the cognitive mechanisms giving rise to religious belief, as opposed to those involved in, for example, perception, are prone to substantial error. The central such suggested mechanism is called an "agency detection device." Such a device, so the thought goes, would probably deliver many false positives; Stewart Guthrie dubs this device "the Hypersensitive Agency Detection Device" (HADD).[20] This device is hypersensitive, he says, because, while little is lost by a false positive, a false negative can be catastrophic. You glimpse a tiger apparently looking at you from a distance of fifty yards; failure to see it as an agent (looking for lunch, perhaps) can be disastrous. By comparison, attributing agency to trees or clouds or the moon is mistaken, but much more benign. ("Better safe than sorry" shows up in nearly all these accounts.)

But of course the fact that belief in supernatural agents arises from HADD (if indeed that is a fact) doesn't even tend to show that such belief is among the false positives; that same device is also responsible for many true beliefs, for example, the belief that there are other minds, other people.[21] Presumably no one would argue that belief in other minds is dubious or irrational or intellectually second rate on the grounds that it has been produced by an agent detection device that sometimes produces false positives. So merely finding or positing a source of religious belief, as with HADD, does nothing to discredit such belief, and neither does pointing out that the source in question delivers false positives.[22] So far we don't have conflict.

20. Guthrie, *Faces in the Clouds: A New Theory of Religion* (New York: Oxford University Press, 1993).
21. See Justin Barrett, "Is the Spell Really Broken?" *Theology and Science*, vol. 5, no. 1, March 2007, p. 69.
22. But wouldn't this at least show that religious beliefs lack *warrant*, that property enough of which is what separates knowledge from mere true belief? (See my *Warranted Christian Belief* (New York: Oxford University Press, 2000), chapter 5.) Not just as such; after all, even if belief in other minds originates in HADD, we do presumably know that there are other people. What counts here is not the global reliability of the faculty or cognitive mechanism in question, but its reliability in the relevant circumstances.

Consider the early Stark proposal—that religion is a spandrel of rational thought and is devoted to the pursuit of nonexistent goods by way of negotiation with nonexistent supernatural agents. This proposal is inconsistent with Christian thought or commitment just because it declares these goods and agents nonexistent. But wouldn't there be another theory, perhaps just as good and even empirically equivalent to Stark's, that was noncommittal on the existence or nonexistence of these goods? And anyway, would one really want to say that it was part of *science*—part of a scientific theory—to declare these goods nonexistent? Suppose I propose as a theory the conjunction of Newton's laws and atheism: have I succeeded in producing a scientific theory inconsistent with theism? Hardly. So delete the offending bit from Stark's theory and call the result "Stark-minus": would Stark-minus be inconsistent with Christian belief? Stark-minus is something like the claim that (a) religion involves the pursuit of certain kinds of ends or goods—salvation, eternal life, and the like—by way of negotiating with alleged supernatural beings, and (b) that it arises as a kind of byproduct or spandrel of the evolution of the capacity for rational thought. Is *that* theory incompatible with Christian thought? Not obviously.

Or consider Wilson and Ruse. According to their theory, there is really no such thing as objective moral obligation, but it is adaptive for people to think that there is. So think about (Wilson and Ruse)-minus, which is the theory that results from theirs when we delete the bit according to which in reality there is no such thing as objective moral obligation. The resulting theory says only that morality—that is, belief in an objective obligation to treat others the way we would like them to treat us, together with the resulting tendency to behave in accordance with this belief to at least some extent—the theory says only that this phenomenon is adaptive at the group level and has become ubiquitous among human beings by way of group selection. Is that incompatible with Christian belief? Again, not obviously; it

adds little to the obvious claim that morality is a civil good, a claim going back to ancient Rome. (Of course it does add the thesis that morality has come to be by way of group selection; perhaps you don't think that's a small addition.) Similarly for Wilson-minus, the theory that results from David Sloan Wilson's theory by deleting the idea that the beliefs involved in religion are fictitious. Wilson-minus, fundamentally, is the theory that religion arises or at least becomes ubiquitous among human beings by way of group selection, because it is a useful form of social control that involves beliefs of a certain kind. (Thus it pays the same compliment to religion that (Wilson and Ruse)-minus pays to morality.) Is this theory incompatible with Christian belief? Again, not obviously. Or consider Atran: he says that religion is "counterfactual," by which I take it he means that religious beliefs are (always? typically?) false. Atran-minus would be the result of deleting the claim that religious belief is false from the rest of his theory: is Atran-minus incompatible with Christian belief? Once more, not obviously.

These theories, therefore, do conflict with religion, but in a merely superficial way. They conflict with religion in the way in which a theory that results from conjoining Newtonian physics with atheism does: that theory conflicts with religion, all right, but it certainly doesn't constitute a serious religion-science conflict.

Here we should briefly pause to ask the following question: what, exactly, *is* a religion/science conflict? As we saw in chapter one, a religion might be itself inconsistent: then of course it will be inconsistent with any scientific theory, but this would be an uninteresting religion/science conflict. Suppose there is a scientific theory that is inconsistent with theistic belief but only a very few scientists endorse it: would that be an example of science/religion conflict? Or does a genuine science/religion conflict require that the theory in question be widely accepted among scientists? But is even that sufficient? What about general relativity and contemporary quantum mechanics? As they stand, they are incompatible with each other; physicists have been trying to develop a theory of quantum gravity to replace them, but so far have been unsuccessful (although

some people look to string theory and its sucessors as a promising source of reconciliation). The deliverances of current science, therefore, contain a contradiction. General relativity and quantum mechanics are inconsistent with each other; hence their conjunction is inconsistent with any religious belief; each is widely accepted among scientists; do we therefore have a religion/science conflict? If so, it would be, again, at best an uninteresting conflict.

So not just any case of explicit religion/science inconsistency is a genuine case of religion/science conflict. Furthermore, conflict can happen in several different ways. For example, a scientific theory might not be explicitly inconsistent with Christian belief, but inconsistent with Christian belief together with propositions that can't sensibly be rejected. A theory might be formally consistent with Christian belief, but still be massively improbable with respect to a set of beliefs or a noetic structure more or less like that of most contemporary Christians, or most contemporary Christians in the Western world (and for that matter, most Christians in the non-Western world). Such Christians will typically believe some propositions F by way of faith, and other propositions R because they are or seem to be deliverances of reason (including memory, perception, rational intuition and so on). A given theory might not be improbable with respect to F and also not improbable with respect to R, but massively improbable with respect to the conjunction of F with R, and hence with respect to a noetic structure that contains both F and R. Such a theory might be so unlikely with respect to such a noetic structure that it wouldn't be a real candidate for belief. An example would be a theory entailing that if human beings have come to be by way of natural selection culling genetic variability, then no rational human being knowingly sacrifices her reproductive prospects in favor of advancing someone else's welfare. This isn't incompatible with F, and perhaps also not incompatible with the deliverances of reason. However, a Christian will think the consequent massively improbable, and might (by virtue of reason) be inclined to accept the antecedent. This theory, then, would be incompatible with the noetic structures of such Christians, even if not logically inconsistent with Christian belief as such. There are still other forms of conflict, as I'll argue later in this chapter.[23]

23. Of course there are conflicts between science and particular religious beliefs that are not part of Christian belief as such: belief in a universal flood, a very young earth, etc.

To look more deeply into this question of conflict or compatibility between Christian belief and these theories from evolutionary psychology, suppose we examine one of them more carefully: David Sloan Wilson's theory of religion.[24] This theory is a so-called "functional interpretation" of religion. Both terms deserve comment. First, Wilson explicitly says many times that his theory is an *interpretation* of religion. This is a bit surprising: one wouldn't think of Newton's laws, for example, or special relativity as interpretations of something or other. What is involved in the theory's being an interpretation? *Understanding* of one sort or another, presumably; the thought is that once you see religion as having the function ascribed to it in the theory, then you understand it, or understand it more deeply. You understand why there is such a thing as religion, why religions arise and persist, and what they are *for*—what their function or purpose is. In the particular case of Wilson's theory, the idea is that religions play an important role in group selection. "Many features of religion, such as the nature of supernatural agents and their relationships with humans, can be explained as adaptations designed to enable human groups to function as adaptive units."[25] (A crucial difference between his theory and those of Atran and Boyer, therefore, is that according to the latter religious belief isn't in fact adaptive.) He aims to "see if the detailed properties of Calvin's Church can be interpreted as adaptation to its environment," and he summarizes his theory as follows:

> I claim that a knowledge of the details [of Calvin's Geneva] clearly supports a group-level functional interpretation of Calvinism. Calvinism is an interlocking system with a purpose:

24. Wilson, *Darwin's Cathedral*, pp. 48ff. See also Robert A. Hinde, *Why Gods Persist* (London: Routledge, 1999), pp. 553ff.
25. Wilson, *Darwin's Cathedral*, p. 51.

> to unify and coordinate a population of people to achieve a common set of goals by collective action. The goals may be difficult to define precisely, but they certainly included what Durkheim referred to as secular utility—the basic goods and services that all people need and want, inside and outside of religion.[26]

The thought is that Calvinism is an interlocking system with a purpose: "to unify and coordinate a population of people to achieve a common set of goals by collective action." This sounds a bit as if he thinks of Calvinism as a project or activity that people undertake in order to achieve a common set of goals, these goals including at least that secular utility of which he speaks. If this were what he means, he would be wrong: Calvin and the other Calvinists weren't (and aren't) embracing Calvinism in order to achieve some kind of secular utility. In fact it is doubtful that Calvinism, or Roman Catholicism, or Christianity or for that matter Judaism or Islam are (wholly) intentional activities in that way at all. Are they human activities undertaken in order to achieve a goal? What is the purpose or aim of being a Calvinist? What is the purpose or aim of believing in God? Well, what is the purpose or aim of believing in other people, or believing that there has been a past? The right answer, one thinks, is that believing in God, like believing in the past or in other people, typically doesn't have any purpose or aim at all. It isn't that you believe in God or other people in order to achieve some end or other. You might as well ask me what my purpose or aim is in believing that I live in Michigan or that 7+5 = 12. In one sense these are intentional activities; but they are not undertaken in order to achieve some end or other.

You may reply that there is more to Christianity in general and Calvinism in particular than holding beliefs. This is certainly true:

26. Wilson, *Darwin's Cathedral*, pp. 91, 118.

there is also love of God, and prayer, and worship, and ritual, ceremony and liturgy, for example. These are activities one intentionally undertakes. But again, it's not clear that there is some *purpose* for the sake of which one undertakes to love God: you love God because he is attractive, such as to attract or compel love. Christians pray because it seems the right thing to do, or because they are instructed to pray, and how to pray, by Jesus Christ. The same holds for worship. When worship is going properly it isn't something done in order to achieve some end outside itself: it is much more spontaneous and immediate than that, and you participate just because it seems right and appropriate. (Of course you *might* engage in worship to please your parents or spouse or children: but then in a case like that it *isn't* going properly.) This is a complex subject, and now is not the time to go into it. What is clear, however, is that there isn't any goal or purpose or end involved, typically, in accepting the central tenets of Calvinism or Christianity, and even if there is a purpose or goal or end involved in worship and prayer, it most certainly is not the achievement of the secular goods Wilson mentions.

But Wilson isn't really proposing that Calvinists themselves engage in the practice of Calvinism in order to achieve those goals. This practice has goals, all right, but they aren't the goals or purposes of the people engaged in the practice. It is rather that the aims or goals are provided, somehow, by evolution. And of course it isn't that these aims or goals are those of evolution, or natural selection; as Wilson is thinking of it, those processes don't have any aims or goals at all and aren't aiming at the actualization of some state of affairs. Still, the idea is that some of the structures and processes that result from natural selection do have purposes, purposes they acquire from their roles in maximizing fitness. The ultimate purpose of the heart, he no doubt thinks, is to enhance or maximize fitness; its proximate purpose is to pump blood (and pump it in a certain way), and the idea is that it fulfills the former purpose by fulfilling that latter. The proximate purpose

of the immune system is to overcome disease; this purpose is in the service of its ultimate purpose of maximizing fitness. Whether one can really speak of purpose and proper function for organs such as the heart or liver or brain absent a designer and outside the context of theism is of course a matter of dispute; I say you can't.[27] But this isn't the place to enter that discussion.

So let's suppose that a heart or a liver, and also an activity like a religion can have a purpose conferred upon it by natural selection, even if God is not orchestrating and guiding that process. According to Wilson, the purpose of a religion, at least in the case of Calvinism, is "to unify and coordinate a population of people." That isn't a purpose endorsed by those who practice the religion; still, he thinks, that is its purpose. Here it is instructive to compare Wilson's views on religion with those of that great master of suspicion, Sigmund Freud. On Freud's view, religion (and here he's thinking especially of theistic religions) is an illusion, in his technical sense. This sense is not such as to entail the falsehood of theistic belief, although in fact Freud thinks theism is false: there is no such person as God. Still, illusions have their uses and indeed their functions. The function or purpose of religious belief is really to enable believers to carry on in this cold and hostile or at any rate indifferent world in which we find ourselves. The idea is that theistic belief arises from a psychological mechanism Freud calls "wish-fulfillment"; the wish in this case is father, not to the deed, but to the belief.[28] Nature rises up against us, cold, pitiless, implacable, blind to our needs and desires. She delivers hurt, fear, pain, anxiety, suffering; and in the end she demands our death. Paralyzed and appalled, we invent (unconsciously, of course) a father in heaven who exceeds our earthly fathers as much in goodness and

27. See chapter 11 of my *Warrant and Proper Function* (New York: Oxford University Press, 1993).
28. And in such a way that it (or its deliverances) rather resembles Calvin's *sensus divinitatis*; see *Moses and Monotheism*, pp. 167ff.

benevolence as in power and knowledge. The alternative would be to sink into depression, stupor, paralysis, and finally death.

This illusion enables us to carry on and survive: therefore it contributes to our fitness. Is this Freudian claim incompatible with Christian belief? Could I accept Christian belief and also accept Freud's explanation or account of it? Well, maybe. For it is at least possible that God gets us to be aware of him by way of a mechanism like wish-fulfillment. According to Augustine, "Our hearts are restless till they rest in you, O God." But then it might be that one way God induces awareness of himself in us is through a process of wish-fulfillment: we want so much to be in God's presence, we want so very much to feel his love, to know that we are loved by the first being of the universe, that we simply come to believe this. I don't say that's in fact the way things go; I say only that it is possible and not incompatible with Christian belief.

There is more to Freud's account, however, than just the idea that we come to believe in God by way of wish-fulfillment; if that were all he thinks, there would be no reason to call theistic belief an *illusion*. What more does Freud say here? That this process of wish-fulfillment isn't *reality oriented*, as we might say; it is this that makes theistic belief an illusion. We human beings display a large number of belief-producing processes or faculties or mechanisms. There is perception, memory, a priori intuition, credulity, induction, and much else. We ordinarily think these faculties or processes are aimed at the production of true belief: that is what they are *for*, and that is their purpose or function. There are some cognitive processes, however, that are not aimed at the production of true belief, but at some other desideratum. Someone may remember a painful experience as less painful than it actually was. According to John 16:21, "a woman giving birth to a child has pain because her time has come; but when her baby is born she forgets the anguish because of her joy that a child is born into the world." You may continue to believe in your friend's honesty

long after evidence and cool, objective judgment would have dictated a reluctant change of mind. I may believe that I will recover from a dread disease much more strongly than is warranted by the statistics of which I am aware. A mountain climber, faced with a life or death situation, might believe more strongly than his evidence warrants that he can leap the crevasse before him.

In all of these cases, there is no cognitive dysfunction or failure to function properly; but the processes in question don't seem to have as their functions the production of true beliefs. Rather, they produce beliefs that in the context are useful in one way or another. And exactly this is the way things stand with Freud's explanation: an essential part of his account of theistic belief is that it is not produced by truth-aimed cognitive processes, but by a process with a different sort of function. At this point the Christian or any serious theist will disagree with him. The serious theist will think that God has created us in such a way that we come to know him; and the function of the cognitive processes, whatever they are, that ordinarily produce belief in God in us is to provide us with true belief. So even if she agrees with Freud that theistic belief arises from wish-fulfillment, she will think that this particular instance of wish-fulfillment is truth-aimed; it is God's way of getting us to see that he is in fact present and in fact cares for us. At this point she will have to disagree with Freud.

Something similar goes for David Sloan Wilson. He holds that the purpose or function of Calvinism and Christianity generally is to enhance fitness; a group with a religion of that sort will do well in competition with groups without any such religion (or anything similar). And specifically religious *belief* plays an important role here. The role of such belief is not to reflect reality, he says, but to play a part in the production of what religion produces. As he says: "our challenge is to interpret the concept of God and his relationship with people as an elaborate belief system designed to motivate the behaviors listed."

In a very interesting passage Wilson proposes that religious belief isn't reality oriented, but, unlike Freud (and most of those who write on the scientific study of religion), goes on to defend it. The passage is worth quoting in full:

> In the first place, much religious belief is not detached from reality. . . . Rather, it is intimately connected to reality by motivating behaviors that are adaptive in the real world—an awesome achievement when we appreciate the complexity that is required to become connected in this practical sense. It is true that many religious beliefs are false as literal description of the real world, but this merely forces us to recognize two forms of realism: a factual realism based on literal correspondence and a practical realism based on behavioral adaptiveness.
>
> In the second place, much religious belief does not represent a form of mental weakness but rather the healthy functioning of the biologically and culturally well-adapted mind. . . . Adaptation is the gold standard against which rationality must be judged, along with all other forms of thought. Evolutionary biologists should be especially quick to grasp this point because they appreciate that the well-adapted mind is ultimately an organ of survival and reproduction. . . . factual realists detached from practical reality were not among our ancestors.[29]

This account of religion, then, is like Freud's in that, like Freud, Wilson sees the cognitive processes that produce religious belief as not aimed at the production of true belief, but at belief that is adaptive by way of motivating those adaptive behaviors. Religious belief in general and Christian and Calvinistic belief in particular originate in

29. Wilson, *Darwin's Cathedral*, p. 228.

belief-producing processes that are aimed, not at the production of true belief, but at the production of belief that will motivate that behavior. And here someone who accepts Christian belief will be forced to demur, just as with Freud. For if Christian belief is in fact true, as, naturally enough, the Christian will think, it will be produced in us by cognitive processes that God has designed with the end in view of enabling us to see the truth of "the great things of the Gospel" (as Jonathan Edwards calls them). She will no doubt think that these processes typically involve what Calvin calls "the internal witness (or testimony) of the Holy Spirit" and what Aquinas calls "the internal instigation of the Holy Spirit." And these processes, so the Christian thinks, will then be truth-aimed: they are aimed at enabling us to form these true beliefs about what God has done and about the way of salvation. So there is indeed a conflict between Wilson's theory of religion and Christian belief.[30]

III HISTORICAL BIBLICAL CRITICISM

A. *Traditional Biblical Commentary*

We turn now to a second area of conflict between Christian belief and science: historical Biblical criticism (HBC, as I'll call it). There are striking parallels, in this regard, between HBC and evolutionary psychology.[31] To see some of these parallels, we must begin with traditional Biblical commentary. Now classical Christians take the Bible

30. Tom Crisp points out (private communication) that (again, as with Freud) there is no incompatibility between religious belief and the thought that it arises in the way Wilson says it does. Clearly God, if he chose, could use the process of group selection to bring it that people are aware of him and in a position to worship him.

31. Here I can be brief; for a much fuller account of HBC, see chapter 12 of my *Warranted Christian Belief*.

to be authoritative in one way or another. That is because they think of the Bible as a special word from the Lord; as they see it, God is the principal and ultimate author of the Bible. Of course the Bible is also a library, each of its books has a human author. But God has used these authors in such a way that what they write has the divine stamp of approval; hence the Bible—the whole Bible—is divinely inspired in such a way that its principal author is God. The Bible is a library, but it is also like a single book in that it has a single principal author. This is the source and warrant for the proposal that we can "interpret scripture with scripture"; if we find a certain obscurity in one of the epistles of Paul, for example, we can look for light, not just elsewhere in that epistle or to other writings of Paul, but also to the epistles of John and the Gospels. And the chief function of the Bible is to disclose to mankind God's gospel—the good news of salvation through the life and death and resurrection of Jesus Christ, himself both a human being but also the divine son of God, the second person of the trinity.

Now what the Lord teaches is of course trustworthy; therefore this entire book, so Christians think, is authoritative. Many questions arise: just how does this work? Just how does this inspiration happen? Just how can it be said that the Bible is divine discourse?[32] This is not the place to enter those discussions. But even if the entire Bible is a word from the Lord, many parts of it are at best difficult to understand. It isn't always easy to tell what the Lord is teaching in a given passage. What he is teaching is certainly true and to be accepted; but it may be hard indeed to see what he *is* teaching. In Colossians 1:24, for example, we read: "Now I rejoice in what was suffered for you, and I fill up in my flesh what is still lacking in regard to Christ's afflictions, for the sake of his body which is the church." But what could be

32. See Nicholas Wolterstorff, *Divine Discourse: Philosophical Reflections on the Claim that God Speaks* (Cambridge: Cambridge University Press, 1995).